

Data Types & the Type System

Primitives · typeof · Coercion · String Methods

HOW TO USE THIS DOCUMENT

- During class** → Keep Part 1 open. Follow the code on your instructor's screen.
- In the 'think it' rows: predict the output BEFORE it runs.
- After class** → Use Part 2 (Visual Summary) to review and remember the concepts.

TODAY'S TOPICS

Topic 1 7 Primitive Data Types

Topic 2 The typeof Operator

Topic 3 Implicit Type Coercion

Topic 4 Explicit Type Conversion

Topic 5 Common String Methods

PART 1 — LIVE CODE (follow along)

Topic 1

7 Primitive Data Types

string · number · boolean · undefined · null · symbol · BigInt

VISUAL REFERENCE

String

"Hello"

"string"

Number

42 / 3.14

"number"

Boolean

true / false

"boolean"

Undefined

undefined

"undefined"

Null

null

"object" ■■

Symbol

Symbol()

"symbol"

BigInt

...991n

"bigint"

undefined = JS set it (no value given) | null = YOU set it (intentionally empty)

FOLLOW ALONG — type this with your instructor

— ■ Predict what console.log prints before your instructor runs the code.

```
// ■■ The 7 Primitive Types ■■  
  
const name      = "Priya";  
const age       = 21;  
const score     = 98.5;  
const isPassed  = true;  
let city;  
const selected  = null;  
const id        = Symbol();  
const bigNumber = 9007199254740991n;
```

Output →

Topic 1

Topic 2

The typeof Operator

check a value's type at any time

VISUAL REFERENCE

```
typeof anyValue → "string" / "number" / "boolean" / "undefined" / "object" / "symbol" / "bigint"
```

FOLLOW ALONG

```
console.log(typeof "Hello");
```



your answer

```
console.log(typeof 42);
```



your answer

```
console.log(typeof true);
```



your answer

```
console.log(typeof undefined);
```



your answer

```
console.log(typeof null);
```



your answer

```
console.log(typeof Symbol());
```



your answer

■ ■ typeof null returns "object" — this is a famous JS bug from 1995. Memorise it.

Topic 3

JavaScript Type Coercion

JS silently converting types

VISUAL REFERENCE

Rule 1 — + with a String

→ JS JOINS them (string concat), NOT adds

```
5 + "3" = "53" ■
```

Rule 2 — - * / with a String

→ JS converts String to Number (math ops only)

```
"10" - 4 = 6 ■
```

FOLLOW ALONG — guess before it runs!

```
console.log(5 + "3");
```



your answer

```
console.log("5" + 3);
```



your answer

```
console.log("10" - 4);
```



your answer

```
console.log("abc" - 1);
```



your answer

```
console.log(true + 1);
```



your answer

```
console.log(false + 1);
```



your answer

Topic 4

Explicit Type Conversion

Number() String() Boolean() parseInt()

VISUAL REFERENCE

Number()

To number

"42"42

"abc"NaN

String()

To string

42"42"

true"true"

Boolean()

To boolean

0false

""false

6 FALSY values (Boolean returns false):0""nullundefinedNaNfalse

FOLLOW ALONG

console.log(Number("42"));

your answer

console.log(Number("hello"));

your answer

console.log(Number(true));

your answer

console.log(parseInt("42px"));

your answer

console.log(String(99));

your answer

console.log(Boolean(0));

your answer

console.log(Boolean("0"));

your answer

Topic 5

Common String Methods

.length · .trim() · .toUpperCase() · .includes() · .slice()

VISUAL REFERENCE

.length

"Hello".length

FOLLOW ALONG

.toUpperCase()

"hello".toUpperCase()

→ ?

.toLowerCase()

"HELLO".toLowerCase()

→ ?

.trim()

" hi ".trim()

→ ?

.includes(val)

"Hello JS".includes("JS")

→ ?

.slice(s, e)

"Hello".slice(0, 3)

→ ?

const greeting = " Hello, JavaScript! ";

console.log(greeting.length);

your answer

console.log(greeting.trim());

your answer

console.log(greeting.trim().toUpperCase());

your answer

console.log(greeting.includes("JavaScript"));

your answer

PART 2 — VISUAL SUMMARY (review after class)

typeof RESULTS

"Hello"	"string"
42 / 3.14	"number"
true/false	"boolean"
undefined	"undefined"
null	"object" ■■
Symbol()	"symbol"
99n	"bigint"

COERCION RULES

5 + "3"	→ "53"	■ + with string = JOIN
"5" + 3	→ "53"	■ + with string = JOIN
"10" - 4	→ 6	■ - converts to number
"abc" - 1	→ NaN	■■ bad math = NaN
true + 1	→ 2	true = 1
false + 1	→ 1	false = 0

EXPLICIT CONVERSION

Number("42")	42	Number("abc")	NaN
Number(true)	1	Number(false)	0
parseInt("42px")	42	String(42)	"42"
Boolean(0)	false ■	Boolean("hi")	true ■
Boolean("")	false ■	Boolean("0")	true ■ ■

STRING METHODS

.length "Hi".length	→ 2	.trim() " hi ".trim()	→ "hi"
.toUpperCase() "hello".toUpperCase()	→ "HELLO"	.includes(val) "Hello JS".includes("JS")	→ true
.slice(s, e) "Hello".slice(0, 3)	→ "Hel"	chaining " hi ".trim().toUpperCase()	→ "HI"